

PROJECT NAME: AUTOMATED NETWORK REQUEST MANAGEMENT SYSTEM

1. INTRODUCTION

1.1 Project Overview

The Automated Network Request Management System is a cloud-based application designed to modernize the process of handling network service requests within an organization. Developed on the ServiceNow platform, the system replaces traditional manual methods such as emails and paper-based tracking with a centralized self-service solution. Employees can easily submit requests for network connections, device provisioning, and office relocations while the platform manages approvals, request tracking, and record maintenance automatically.

1.2 Purpose

The purpose of this project is to simplify the network request process by eliminating manual intervention and reducing operational delays. Through automated workflows, accurate data collection, and real-time tracking, the system enhances service efficiency, maintains data integrity, and provides complete visibility throughout the request lifecycle.

2. IDEATION PHASE

2.1 Problem Statement

Organizations often rely on manual communication channels to process network-related requests. These traditional approaches frequently result in incomplete request information, approval delays, repeated follow-ups, and data entry mistakes. Such challenges slow down office relocations, network provisioning activities, and overall business operations, ultimately affecting employee productivity.

2.2 Empathy Map Canvas

To better understand user expectations and challenges, an empathy analysis was performed for employees requesting network services.

- **User Says:** "I'm unable to know the progress of my network request."

- **User Thinks:** "The approval process should be faster and more transparent."
- **User Does:** Frequently follows up with the IT team through emails or calls for updates.
- **User Feels:** Frustrated by delays and uncertain due to the lack of request visibility.

2.3 Brainstorming

The brainstorming session identified several possible improvements to automate the existing process.

- Design a centralized Service Portal for submitting network requests.
- Implement automated approval workflows using Flow Designer.
- Store approved requests securely in a dedicated custom database for future reference.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey consists of four important stages.

1. **Discovery:** Employees access the Service Portal to submit their network requests.
2. **Configuration:** User information is automatically captured while additional fields appear based on the selected request type.
3. **Approval Pending:** The request is forwarded to the reporting manager, and users can monitor the approval status.
4. **Fulfillment:** Once approved, the request is stored in the custom database and a confirmation is provided to the employee.

3.2 Solution Requirement

Functional Requirements (FR)

The proposed system should:

- Allow employees to submit requests through the Service Portal.
- Automatically retrieve employee profile details.
- Display dynamic form fields based on request selection.
- Route requests for manager approval automatically.
- Generate notifications during different workflow stages.
- Save approved request information into the custom database.

Non-Functional Requirements (NFR)

The solution should provide:

- A user-friendly and responsive interface.
- Secure access using role-based permissions.
- Reliable and accurate request processing.
- High performance with minimal response time.
- Scalable architecture for future enhancements.

3.3 Data Flow Diagram

The application processes requests through the following workflow:

Service Portal Request



Client-side Validation



Flow Designer Workflow



Manager Approval



Custom Database (u_network_database)

3.4 Technology Stack

- **Frontend Platform:** ServiceNow Service Portal, HTML, CSS, JavaScript
- **Process Automation:** Flow Designer, UI Policies, Client Scripts
- **Backend Database:** Custom Table (u_network_database)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed design directly addresses the issues identified in the existing manual process. Automated data validation reduces input errors, workflow automation speeds up approvals, and real-time status tracking improves transparency for both employees and administrators.

4.2 Proposed Solution

The solution introduces a centralized Service Portal where employees can submit network-related requests. Flow Designer automates the approval process, while approved requests are securely stored in a custom database. This architecture minimizes manual work, improves operational efficiency, and ensures accurate record management.

4.3 Solution Architecture

The solution follows a structured multi-layer architecture consisting of four stages:

1. Portal Submission
2. Flow Orchestration Engine
3. Multi-Stage Approval Gate
4. Secure Injection & Database Write

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was developed using the Agile Scrum methodology and completed across two planned sprints. Development tasks were estimated using Story Points to measure effort and complexity.

Sprint-1: UI Logic & Intake Architecture (7 Story Points)

- USN-1 (2 Points): Configure automatic employee profile retrieval.
- USN-2 (3 Points): Develop dynamic form visibility using UI Policies.
- USN-3 (2 Points): Build the frontend request submission process.

Sprint-2: Governance, Security & Ledger Commits (12 Story Points)

- USN-4 (2 Points): Configure automated manager approvals and notifications.
- USN-5 (5 Points): Develop server-side scripts to store approved requests.
- USN-6 (5 Points): Create dashboards for monitoring request activities.

Velocity Metric Summary

- Total Story Points Completed: 19
- Average Velocity: 9.5 Story Points per Sprint

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

User Acceptance Testing (UAT) and functional validation were carried out to verify the overall performance of the application. The testing focused on validating form functionality, workflow execution, approval processing, email notifications, security rules, and database updates.

Defect Resolution Tracking Matrix

- Critical workflow issues were identified, corrected, and successfully retested.
- Minor interface alignment and display issues were resolved to improve user experience.

Test Suite Pass-Rate Matrix

- Service Portal Form – 100% Passed
- Flow Designer Workflow – 100% Passed
- Access Control Lists – 100% Passed
- Database Integrity – 100% Passed
- Exception Handling – 100% Passed

Overall, all planned test cases were completed successfully without any unresolved issues.

7. RESULTS

7.1 Output Screenshots

The completed application successfully demonstrates the complete network request lifecycle.

The results include:

- Service Portal Request Form
- Order Confirmation Page
- Approval Workflow
- Email Notification
- Requested Item Record
- Database Record
- Dashboard Reports

8. ADVANTAGES & DISADVANTAGES

Advantages

- Eliminates manual request processing.
- Improves request accuracy through automation.

- Provides real-time request tracking.
- Reduces approval delays.
- Enhances data security and consistency.
- Improves overall user experience.

Disadvantages

- Dependent on ServiceNow platform availability.
- Initial configuration requires development effort.
- Future enhancements may require additional customization.

9. CONCLUSION

The Automated Network Request Management System successfully automates the complete lifecycle of network service requests. By integrating Service Portal, Flow Designer, UI Policies, and custom database tables, the application reduces manual effort, improves processing speed, enhances transparency, and maintains accurate data management. The implemented solution offers a reliable and efficient approach for managing network requests within an organization.

10. FUTURE SCOPE

Future enhancements can include integrating Machine Learning and Predictive Intelligence to automate request categorization and assignment. Additional improvements may involve intelligent resource allocation, mobile accessibility, advanced reporting dashboards, and integration with enterprise asset management systems to further optimize network request management.

11. APPENDIX

GitHub & Project Demo Link:

□ Repository: [manosika/Automated-Network-Request-Management-in-ServiceNow](https://github.com/manosika/Automated-Network-Request-Management-in-ServiceNow)